

Benchmarking Vision-Language Models for Robotic Perception and Planning

From Object Grounding to Hierarchical Task Execution

Louis Jadot
University of Waikato
louis.jadot04@gmail.com

May 27, 2026

Abstract

We present a five-stage benchmark evaluating Vision-Language Models (VLMs) for robot perception, grasp reasoning, and sequential task planning. Five state-of-the-art VLMs—GPT-5.4 Mini, Gemini 3.1 Flash Image, Qwen 3.5 9B, Seed 2.0 Lite, and Ministral 8B—are evaluated through a progressive set of robotic manipulation tasks requiring structured JSON outputs.

Test 1 evaluates isolated single-object grasping. **Test 2** introduces multi-object scenes requiring target localization. **Test 3** increases scene clutter and enforces strict spatial grounding. **Test 4** evaluates grounded multi-step execution planning in real-world domestic scenes. **Test 5** evaluates high-level task decomposition and autonomous planning under varying levels of task ambiguity.

Tests 1–3 use a $[0, 5]$ scoring scale and show that GPT-5.4 Mini consistently achieves the strongest perception and grasping performance ($\mu = 4.88, 4.80, 4.73$), while Seed 2.0 Lite demonstrates remarkable stability across increasing perceptual difficulty. Open-weight models, particularly Ministral 8B, exhibit severe degradation under strict localization constraints.

In Test 4 ($[0, 1]$ scale), Gemini Flash Image achieves the highest execution planning score ($\mu = 0.751$), narrowly surpassing GPT-5.4 Mini ($\mu = 0.745$), while Qwen 3.5 9B shows catastrophic variance ($\sigma = 0.391$). Test 5 confirms the dominance of GPT-5.4 Mini ($\mu = 0.788$) and Gemini Flash Image ($\mu = 0.785$) for hierarchical task decomposition and grounded planning. Across all planning-oriented evaluations, Scene Understanding emerges as the primary bottleneck criterion.

Inter-evaluator agreement in Test 1, recomputed after three independent runs per LLM judge, rises from $r = 0.50$ (single run) to $r = 0.91$ (3-run averaged), and reaches $r = 0.931$ in the structured high-level planning task (Test 5), confirming that both *rubric structure* and *multi-run averaging* are required for reliable LLM-as-judge benchmarks.

1 Introduction

[1, 2]

This paper presents a five-test benchmark that progressively increases task complexity along two axes: *perceptual difficulty* (Tests 1–3) and *planning depth* (Tests 4–5).

Test 1 evaluates isolated single-object grasping. Test 2 introduces multi-object scenes requiring target localization. Test 3 adds clutter and enforces a stricter spatial grounding rubric. Test 4 shifts from perception to grounded *execution planning*, where the model must generate a complete sequence of physically coherent manipulation steps for a real-world scene. Finally, Test 5 evaluates *high-level autonomous planning*: the model must decompose an underspecified task into atomic subtasks before producing a grounded execution strategy.

Together, these five evaluations provide a unified benchmark spanning perception, localization, grasp reasoning, execution planning, and hierarchical task decomposition for robotics-oriented VLMs.

The remainder of this paper is structured as follows. Section 2 reviews relevant work. Section 3 describes the evaluation protocols. Section 4 reports Test 1 (single object). Section 5 reports Test 2 (multi-object). Section 6 reports Test 3 (strict localization). Section 9 reports Test 4 (multi-step execution planning). Section 10 reports Test 5 (high-level autonomous planning). Section 11 provides a cross-test synthesis. Section 12 discusses implications and limitations.

1.1 Contributions

This work makes four primary contributions:

1. We introduce a progressive five-stage benchmark spanning object perception, localization, grasp reasoning, execution planning, and high-level task decomposition.
2. We propose structured scoring protocols for both low-level grasping and high-level robotic planning tasks using dual LLM evaluators, and demonstrate that multi-run averaging is essential for reliable agreement.

3. We provide one of the first comparative analyses of proprietary and open-weight VLMs across increasingly complex robotics-oriented reasoning tasks.
4. We identify scene understanding and spatial grounding as the primary bottlenecks limiting current VLM deployment in robotic manipulation systems.

2 Related Work

Existing VLM benchmarks address either (i) general visual question answering [3, 4, 5], (ii) bounding-box grounding [6], or (iii) embodied policies inside specific robot stacks [7, 8, 9, 10]. None of them jointly evaluate zero-shot grasp reasoning, textual spatial localization without coordinates, full execution planning, and autonomous task decomposition under linguistic ambiguity, using a unified scoring protocol and dual LLM judges. Our benchmark closes exactly this gap and is, to our knowledge, the first to chart the *capability degradation curve* of frontier VLMs as a function of robotics-relevant task complexity.

VLM benchmarks. Standard multi-modal benchmarks such as VQA [3], MMMU [4], and MMBench [5] focus on global visual understanding and question answering. These benchmarks do not assess robot-centric affordance reasoning or structured grasp planning, motivating our task-specific evaluation.

Object localization in VLMs. Grounding-capable models such as Grounding DINO [6] address object localization through bounding-box prediction. Our benchmark evaluates a complementary skill: *textual* localization of an object within a scene as part of a downstream manipulation plan, without requiring spatial coordinates.

Robotic grasping and language. Language-conditioned manipulation has been studied in RT-1 [7], RT-2 [8], and SayCan [9]. PaLM-E [10] and similar embodied models demonstrate VLM integration in closed-loop robotic systems. Our work provides a *task-agnostic* quantitative evaluation of raw VLM capabilities before any robotics fine-tuning.

LLM-as-judge evaluation. We use two LLM judges to score VLM responses, following the LLM-as-judge paradigm [11]. Inter-evaluator agreement is measured via Pearson r , Spearman ρ , and MAE across both evaluators, allowing us to quantify scoring reliability as a function of task structure.

3 Evaluation Protocol

3.1 Models

The five VLMs evaluated across all tests are described in Table 1.

Table 1: VLMs evaluated. “—” = undisclosed parameter count.

Model	Type	Params
GPT-5.4 Mini	Proprietary	—
Gemini 3.1 Flash Image	Proprietary	—
Seed 2.0 Lite	Proprietary	—
Qwen 3.5 9B	Open-weight	9B
Ministral 8B	Open-weight	8B

3.2 Object Dataset

Eight everyday objects are used across Tests 1–3: *blue pen*, *grey mug*, *computer mouse*, *orange balloon*, *blue can*, *USB drive*, *whiteboard eraser*, and *stirrer*. In Test 1 each object is presented in isolation; in Test 2 the same object appears in a multi-object scene alongside visual distractors. In Test 3, the same objects appear in a cluttered scene where some are partially occluded. Tests 4 and 5 use 10 photographs of real domestic environments (table after a meal, kitchen, bathroom, laundry room, sink, etc.).

3.3 Scoring System — Tests 1–3

These tests use an identical five-criterion rubric. Each criterion is scored 0–5; the final score is their arithmetic mean.

(A) Identification. *Test 1*: name match (exact=3, synonym=1.5, wrong=0) plus attribute bonus (+0.5 per valid property, max 2). If wrong name: cap at 2. *Tests 2–3*: the name sub-score is replaced by a *position score* (+1 per valid spatial descriptor, max 3); attribute bonus unchanged. Describing the wrong object yields A=0.

(B) Planning (0–5). Approach 0.5 pt, pre-grasp 0.75 pt, grasp 1.0 pt, lift 0.75 pt (= 3 pts actions); logical order 1 pt + efficiency 1 pt (= 2 pts quality).

(C) Use (0–5). Realism (2 valid uses=2, 1=1), grasp-use link (explicit=2, implicit=1, absent=0), physics plausibility 1 pt.

(D) Grasp (0–5). Type correctness (2 pts), contact zone precision (2 pts), accessibility 1 pt.

(E) Robustness (0–5, rescaled). Hallucination (none=1, minor=0.5, major=0) + logic (perfect=1, minor=0.5, incoherent=0). Raw score $\in [0, 2]$ rescaled: $E_{\text{scaled}} = E_{\text{raw}} \times 2.5$.

Final score. Score = $(A + B + C + D + E_{\text{scaled}})/5$.

3.4 Dual-Evaluator Validation

Each VLM response is independently scored by **GPT-5.2** and **Claude Sonnet 4.6** acting as LLM judges [11]. All reported scores are means over both evaluators. Inter-evaluator agreement is measured by Pearson r , Spearman ρ , and Mean Absolute Error (MAE). In Test 1, three independent runs per evaluator are averaged before correlating, following the hierarchical averaging procedure described in Section 4.

4 Test 1: Single-Object Grasping

4.1 Task

The VLM receives a single image of one object with no scene context and produces a JSON response with five sections: `object_identification` (name + attributes), `grasp`, `planning`, `use`, `robustness`.

4.2 Multi-Run Averaging Protocol

To address the low inter-evaluator agreement observed in the original single-run evaluation ($r = 0.50$), two additional independent runs were collected per LLM judge, yielding $3 \text{ runs} \times 2 \text{ evaluators} \times 5 \text{ models} \times 8 \text{ objects} = 240$ rated responses. All Test 1 scores are computed by first averaging each evaluator’s 3 runs at the (model, object) level, then averaging the two evaluators—preserving the original dual-evaluator protocol while substantially reducing per-judge noise before correlation.

4.3 Results

Table 2 reports updated subscores with 95% bootstrap confidence intervals (10 000 resamples over 8 objects) and within-evaluator run standard deviation $\bar{\sigma}_{\text{run}}$. GPT-5.4 Mini leads with $\mu = 4.88$ and perfect Planning and Grasp scores (5.00). Gemini 3.1 Flash Image ranks second ($\mu = 4.67$). Ministral 8B is last ($\mu = 3.91$) with the weakest Robustness (2.66) and highest latency (20.2 s).

Table 2: Test 1 — updated mean subscores, final score with 95% bootstrap CI, and within-evaluator run standard deviation. Three runs per evaluator, two evaluators, 8 objects. Best per column in **bold**.

Model	A	B	C	D	E	Final	95% CI	$\bar{\sigma}_{\text{run}}$
GPT-5.4 Mini	4.86	5.00	5.00	5.00	4.53	4.88	[4.80, 4.94]	0.08
Gemini 3.1 Flash	4.87	4.94	5.00	4.80	3.75	4.67	[4.58, 4.75]	0.20
Seed 2.0 Lite	4.72	4.76	4.86	4.65	4.00	4.60	[4.16, 4.85]	0.39
Qwen 3.5 9B	4.03	4.84	4.83	4.59	3.12	4.28	[4.05, 4.52]	0.28
Ministral 8B	3.90	4.52	4.67	3.76	2.66	3.91	[3.68, 4.16]	0.50

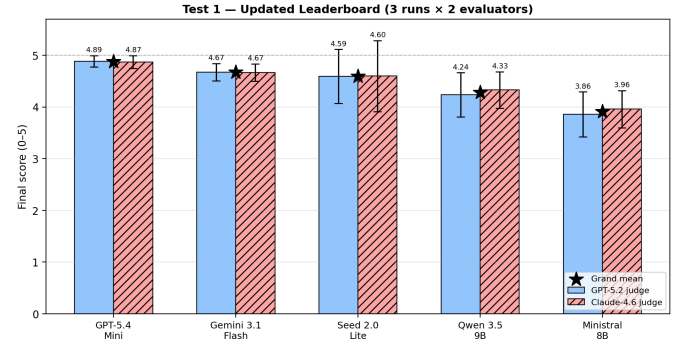


Figure 1: Test 1 — Updated leaderboard ($3 \text{ runs} \times 2 \text{ evaluators}$). Bars: mean over 8 objects and 3 runs, per evaluator; error bars = across-object std. Stars: grand mean (both evaluators averaged).

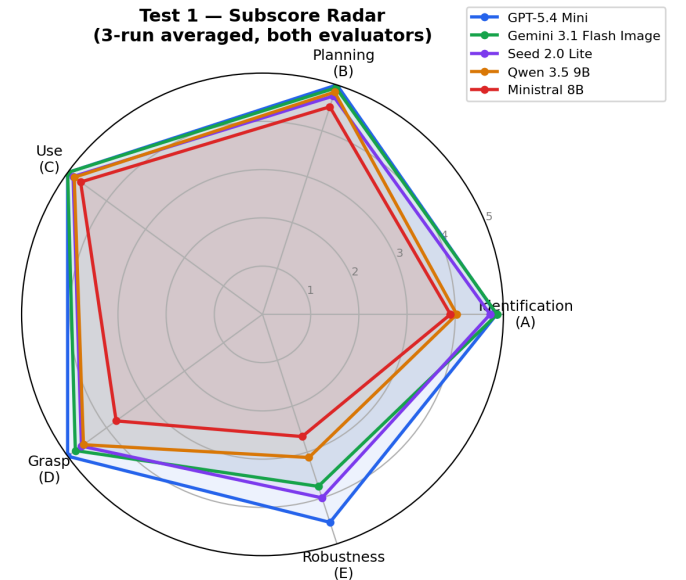


Figure 2: Test 1 — Subscore radar (3-run averaged, both evaluators).

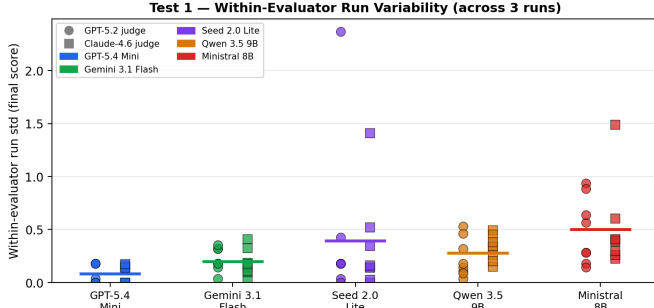


Figure 3: Test 1 — Within-evaluator run variability. Each point is the standard deviation of an evaluator’s three runs on one (model, object) cell. Horizontal bars show per-model means. GPT-5.4 Mini is the most stable ($\bar{\sigma} = 0.08$); Ministral 8B the most volatile ($\bar{\sigma} = 0.50$).

Ranking stability. The relative ordering is preserved (GPT-5.4 Mini > Gemini > Seed > Qwen > Ministral). Seed 2.0 Lite’s widened CI [4.16, 4.85] reveals real run-level instability ($\bar{\sigma}_{\text{run}} = 0.39$) partially masked in the original single-run estimate. Ministral 8B’s mean falls from 4.00 to 3.91, correcting an upward single-run bias.

Inter-evaluator agreement. With three runs per evaluator, agreement strengthens substantially: Pearson $r = 0.91$ (vs. 0.50 originally), Spearman $\rho = 0.89$, MAE = 0.16. The residual gap with Tests 2–3 ($r \approx 0.90$ –0.94) confirms that single-object open-ended evaluation remains inherently more subjective than structured localization tasks—but the earlier “moderate agreement” conclusion was largely an artifact of single-run noise.

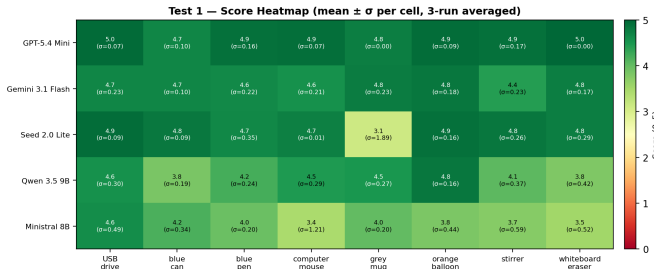


Figure 4: Test 1 — Score heatmap (mean / run σ per cell, 3-run averaged).

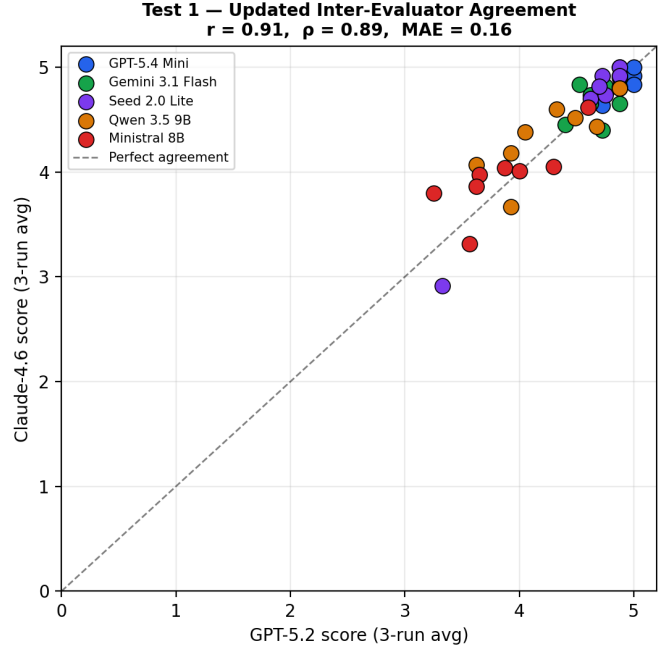


Figure 5: Test 1 — Updated inter-evaluator agreement on 3-run-averaged scores. Each point = one (model, object) cell ($n = 40$). $r = 0.91$, $\rho = 0.89$, MAE= 0.16.

Key observations: (i) Planning and Use are easy for all models (> 4.5); (ii) Robustness is the main differentiator; (iii) multi-run averaging substantially improves inter-evaluator reliability, demonstrating that single-run LLM-as-judge protocols systematically underestimate agreement in open-ended evaluation settings.

The single-object setting establishes a near-ceiling for the proprietary models. We next stress-test whether this ceiling survives the introduction of distractor objects requiring textual localization.

5 Test 2: Multi-Object Scene Grasping

5.1 Task

The VLM receives an image with *multiple* objects and a textual hint specifying the target (e.g., “blue pen”). The model must first *localize* the target in the scene (new `position_in_scene` field in the JSON) before reasoning about grasp and use. The Identification criterion (A) thus evaluates localization accuracy rather than pure name recall.

5.2 Results

Table 3 reports Test 2 results. The top-2 ranking is preserved (GPT-5.4 Mini: $\mu = 4.80$; Seed 2.0 Lite: $\mu = 4.66$) while the bottom changes significantly.

Table 3: Test 2 — mean subscores and final score (both evaluators, 8 objects). Best per column in **bold**. * = includes one JSON failure (score 0).

Model	A	B	C	D	E	Final	RJ
GPT-5.4 Mini	4.69	4.97	5.00	4.72	4.61	4.80	15.
Seed 2.0 Lite	4.62	4.97	5.00	4.86	3.87	4.66	15.
Gemini 3.1 Flash	4.59	4.83	4.89	4.75	3.79	4.57	17.
Ministral 8B	2.56	4.73	4.81	4.53	2.58	3.84	17.
Qwen 3.5 9B*	2.91	4.28	3.91	3.94	2.58	3.52	15.

Localization failure in open-weight models. The most striking result is the collapse of Identification scores for Ministral 8B (A=2.56) and Qwen 3.5 9B (A=2.91), compared to scores above 3.9 and 4.0 in Test 1 respectively. Both models struggle to reliably locate the target object within the scene, confirming that multi-object scenes pose a specific challenge for smaller open-weight VLMs.

JSON parse failure. Qwen 3.5 9B produces an invalid JSON response for the *stirrer* object across *both* evaluators (score = 0.00), a failure mode not observed in Test 1. This suggests that the added complexity of scene description in the prompt can push the model toward formatting errors, a critical reliability concern for production robotics systems. If the invalid JSON generation is excluded from evaluation, Qwen 3.5 9B would achieve adjusted scores of A=3.33, B=4.89, C=4.47, D=4.50, E=2.95, and a final score of 4.02 instead of 3.52, which would have made it possible to surpass Gemini 3.1 Flash Image.

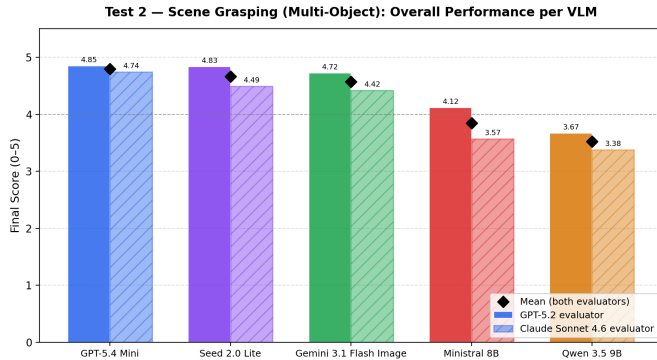


Figure 6: Test 2 — Final scores per VLM split by evaluator. Stars = mean.

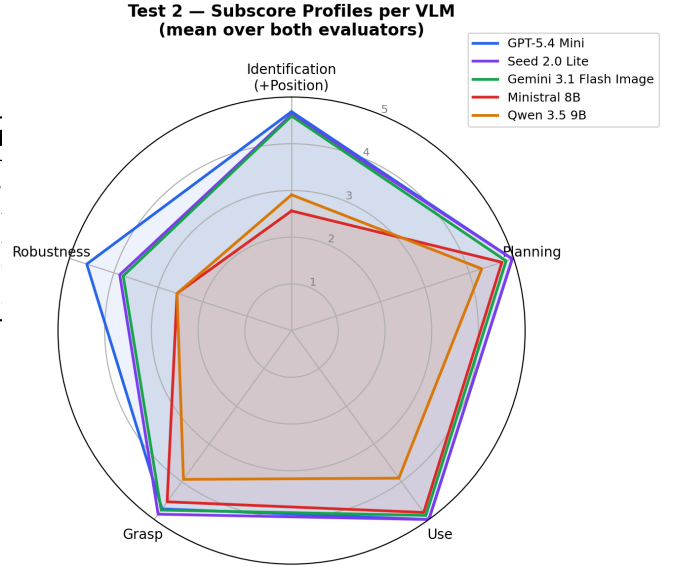


Figure 7: Test 2 — Subscore radar. Note the severe Identification and Robustness dip for Ministral 8B and Qwen 3.5 9B.

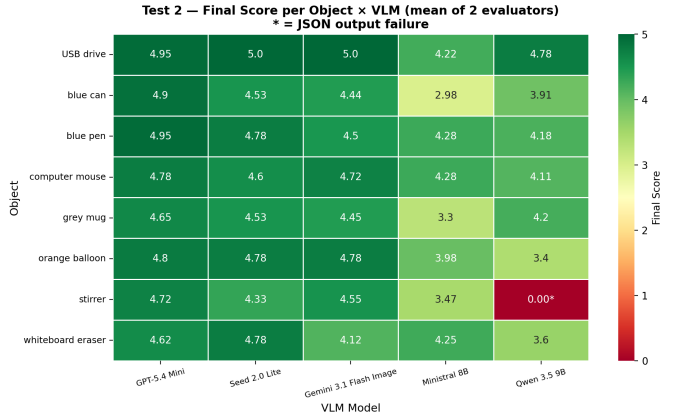


Figure 8: Test 2 — Score heatmap. * denotes the Qwen 3.5 9B JSON failure on *stirrer*.

Inter-evaluator agreement improves. In stark contrast to Test 1 single-run ($r = 0.50$), inter-evaluator agreement in Test 2 is *strong* ($r = 0.94$, MAE= 0.36). Clear localization failures produce unambiguous low scores that both LLM judges assign consistently, reducing the subjectivity inherent in nuanced single-object evaluation.

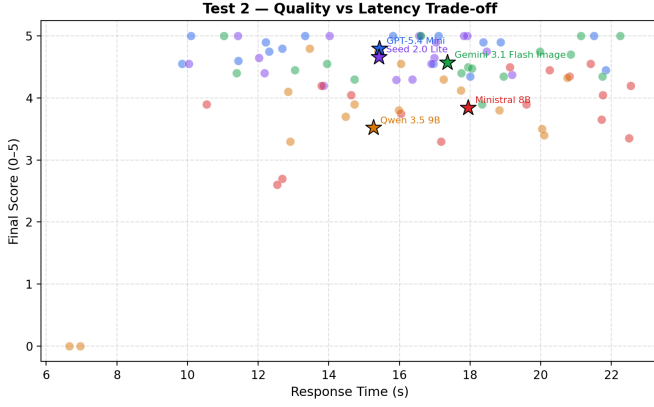


Figure 9: Test 2 — Quality vs latency. GPT-5.4 Mini and Seed 2.0 Lite jointly occupy the Pareto-optimal region.

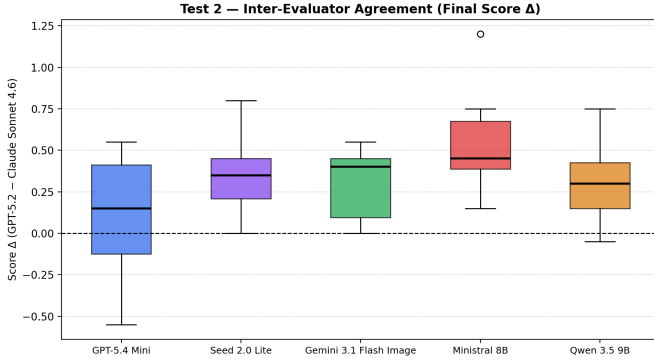


Figure 10: Test 2 — Inter-evaluator agreement. GPT-5.2 systematically scores Ministral 8B higher and gives globally higher scores than Claude Sonnet 4.6.

While multi-object localization separates open- from closed-weight models, the rubric still tolerates approximate positional language. Test 3 tightens this rubric to isolate *precise* spatial grounding.

6 Test 3: Strict Localization Grasping

6.1 Task

Building on the localization failures observed in Test 2, Test 3 uses the same multi-object scene setup but tightens the Identification rubric for cluttered scenes. The `position_in_scene` field is now graded as *+1 per valid spatial descriptor* (orientation, where in the image, nearby objects) up to a maximum of 3 points, replacing the binary correct/approximate/absent scoring of Test 2. Additionally, describing the wrong object now yields a mandatory score of 0 for criterion A (no partial credit). This forces models to provide rich, precise localization rather than vague positional approximations.

6.2 Results

Table 4 shows that the top two models remain stable while the open-weight models diverge sharply. No JSON failures were observed in Test 3.

Table 4: Test 3 — mean subscores and final score (both evaluators, 8 objects). Best per column in **bold**.

Model	A	B	C	D	E	Final	RT
GPT-5.4 Mini	4.66	4.95	5.00	5.00	4.06	4.73	14.7
Seed 2.0 Lite	4.56	4.88	5.00	4.84	3.75	4.61	15.8
Gemini 3.1 Flash	3.88	4.75	4.86	4.78	3.36	4.33	17.9
Qwen 3.5 9B	2.91	4.59	4.81	4.66	2.73	3.93	18.0
Ministral 8B	1.28	4.63	4.75	3.91	2.11	3.33	17.9

Ministral 8B collapses on localization. The most dramatic finding is Ministral 8B’s Identification score of 1.28/5—the lowest recorded across all three tests. Its position sub-score averages just 0.69 out of 3, indicating that the model systematically fails to describe where the target object is in the scene. Planning (4.63) and Use (4.75) remain high, confirming that the model can reason about manipulation in isolation but cannot ground its reasoning in the observed scene.

Qwen 3.5 9B partially recovers. After its Test 2 collapse ($\mu = 3.52$, driven by a JSON failure), Qwen 3.5 9B recovers to 3.93 in Test 3—but remains -0.35 below its Test 1 baseline. Its position sub-score (1.38/3) is weak but functional, unlike Ministral 8B’s.

Gemini 3.1 Flash Image shows unexpected degradation. Unlike GPT-5.4 Mini and Seed 2.0 Lite which lose less than 0.1 points per test, Gemini 3.1 Flash Image drops -0.24 from Test 2 to Test 3. Its Identification score falls from 4.59 to 3.88, suggesting that the stricter localization rubric penalizes its tendency toward approximate positional descriptions.

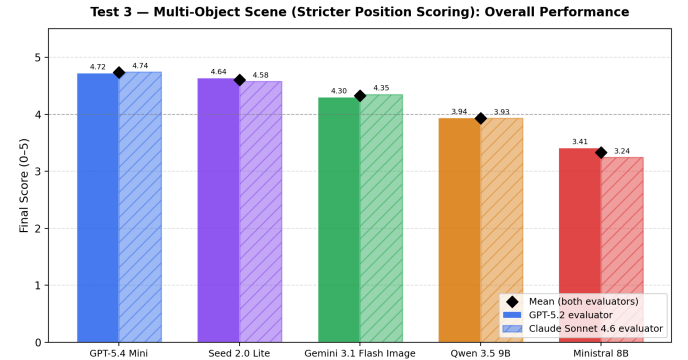


Figure 11: Test 3 — Final scores per VLM split by evaluator. Stars = mean.

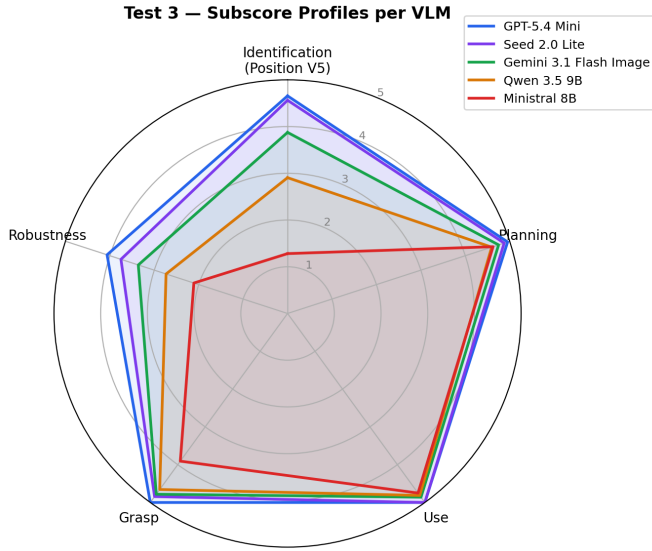


Figure 12: Test 3 — Subscore radar. Minstral 8B’s Identification spike inward reveals its localization failure.

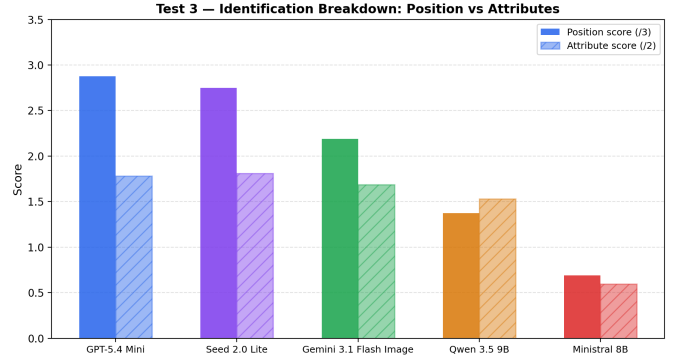


Figure 14: Test 3 — Identification decomposed into Position (solid) and Attribute (hatched) sub-scores. The position gap between proprietary and open-weight models is the primary performance driver.

Inter-evaluator agreement. Test 3 maintains strong agreement ($r = 0.90$, $MAE = 0.19$), confirming the trend established in Test 2: structured localization tasks yield reliable LLM-judge evaluations. The MAE of 0.19 is the lowest across Tests 1–3, suggesting that the stricter rubric leaves less room for evaluator subjectivity.

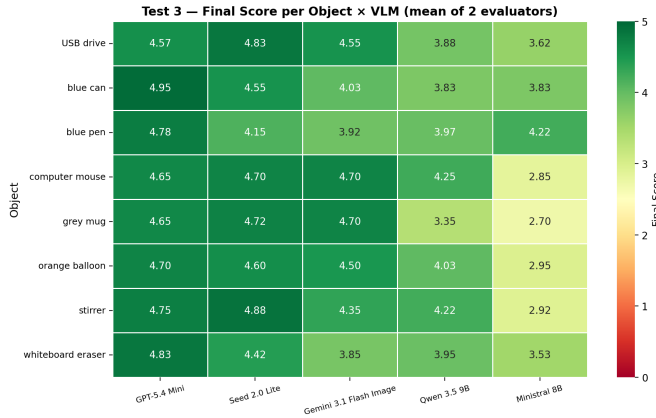


Figure 13: Test 3 — Score heatmap per object $\times$ VLM.

Identification breakdown. Figure 14 separates Position and Attribute sub-scores. The gap between proprietary and open-weight models is almost entirely driven by Position: GPT-5.4 Mini achieves 2.88/3 and Seed 2.0 Lite 2.75/3, while Minstral 8B scores only 0.69/3. Attribute scoring is more uniform (range 0.59–1.81), suggesting that basic visual attribute description is within reach of all evaluated models.

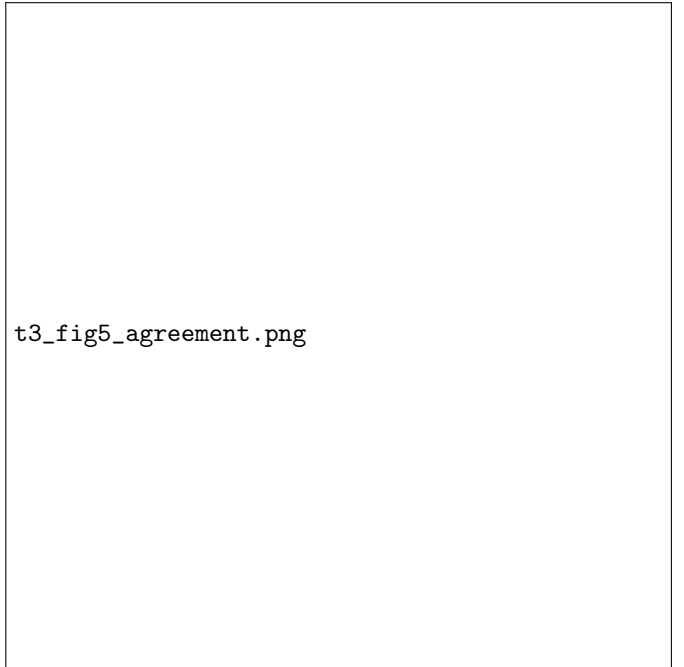


Figure 15: Test 3 — Inter-evaluator agreement. The two LLM judges are tightly aligned, particularly for Minstral 8B’s failures.

Tests 1–3 jointly characterize *static perception*. We now shift to *dynamic reasoning*, where VLMs must produce executable manipulation sequences (Test 4) and ultimately decompose under-specified goals (Test 5).

7 Cross-Test Comparison (Tests 1–3)

Table 5 summarizes final scores across the first three tests.

Table 5: Final score across Tests 1–3 (mean over both evaluators and 8 objects). $\Delta_{12} = T2 - T1$, $\Delta_{23} = T3 - T2$.

Model	T1	T2	T3	Δ_{12}	Δ_{23}	Trend
GPT-5.4 Mini	4.88	4.80	4.73	−0.08	−0.07	Stable
Seed 2.0 Lite	4.60	4.66	4.61	+0.06	−0.05	Stable
Gemini 3.1 Flash	4.67	4.57	4.33	−0.10	−0.24	Declining
Qwen 3.5 9B	4.28	3.52	3.93	−0.76	+0.41	Volatile
Ministral 8B	3.91	3.84	3.33	−0.07	−0.51	Declining

Four distinct behavioral profiles emerge:

Stable-top (GPT-5.4 Mini, Seed 2.0 Lite). Both models lose less than 0.1 points over three increasingly difficult tests. GPT-5.4 Mini shows a near-linear decline (−0.07–0.08 per test step), while Seed 2.0 Lite dips slightly in Test 3 after peaking in Test 2.

Gradually declining (Gemini 3.1 Flash Image). Gemini maintains competitive performance in Tests 1–2 but degrades more significantly in Test 3 (−0.24). Its approximate positional descriptions, tolerated in Test 2, are penalized by the stricter Test 3 rubric.

Volatile (Qwen 3.5 9B). Qwen suffers the largest single-step drop (−0.76, Test 1→2) due to a JSON failure, then partially recovers (+0.41, Test 2→3). This non-monotonic trajectory reflects the fragility of its JSON output generation rather than stable perception capability.

Monotonically declining (Ministral 8B). Ministral 8B declines at every step, with its steepest drop in Test 3 (−0.51). Its Identification score in Test 3 (1.28) is nearly 3.4 points below GPT-5.4 Mini (4.66), a gap entirely attributable to spatial scene description.

	Test 1 (3 runs)	Test 2	Test 3
Pearson r	0.91	0.94	0.90
Spearman ρ	0.89	—	—
MAE	0.16	0.36	0.19

Figure 16: Inter-evaluator agreement across Tests 1–3. Test 1 recomputed after 3-run averaging; agreement now increases monotonically with rubric structure.

8 Discussion (Tests 1–3)

Spatial scene description is the key differentiator. Across all three tests, Planning and Use scores are high for all models (> 4.3). The principal axis of differentiation is the ability to *precisely describe where an object is in a scene*. Test 3 reveals a clear two-tier structure: proprietary models (position score $\approx 2.8/3$) vs. open-weight models (position score $\leq 1.4/3$).

Robustness as the most volatile criterion. The Robustness (E) criterion shows the most variation across models and tests, consistently the weakest for open-weight models and sensitive to prompt complexity.

JSON reliability in complex prompts. Test 2’s invalid JSON generation (Qwen 3.5 9B on *stirrer*) was not reproduced in Test 3 despite the increased task complexity, suggesting that structured-output failures may not depend solely on prompt difficulty.

Evaluator reliability increases with task structure. The progression $r = 0.91 \rightarrow 0.94 \rightarrow 0.90$ (after multi-run averaging in Test 1) confirms that structured, binary evaluation criteria produce far more consistent LLM-judge scores than open-ended qualitative judgments [11]. Crucially, single-run agreement in Test 1 ($r = 0.50$) is not representative of the true inter-judge reliability: three-run averaging improves Pearson r by +0.41, a methodologically important finding for future benchmark designs.

9 Test 4: Multi-Step Execution Planning

9.1 Task and Protocol Differences

Test 4 represents a fundamental shift in the evaluation paradigm. Rather than evaluating *single-object perception*, the VLM must produce a complete **multi-step execution plan** grounded in a real-world scene image. Each model receives: (i) a scene photograph, (ii) an ordered list of 5 manipulation tasks, and must output a valid JSON plan covering scene understanding, step-by-step object selection, grasp strategy, action type, pre/post conditions, and state transitions.

Key differences from Tests 1–3: (i) scores on a $[0, 1]$ scale; (ii) 10 distinct real-world scenes; (iii) each (model, scene) pair is evaluated over **3 independent runs** to measure consistency; (iv) 7 scoring criteria.

This yields 10 scene×5-task combinations evaluated over 3 runs, totaling 150 sequential evaluations per judge.

9.2 Scoring Protocol

OV — Object Validity: visibility (1), class correctness (1), attribute consistency (1). If object not visible $\Rightarrow$ all step scores = 0. **GD — Grasp Definition:** type validity (1), contact zone (1), accessibility (1). **PPC — Physical Properties:** stability (1), fragility (1), manipulability (1). **AE — Action Execution:** action matches task (1), physically possible (1), targets correct object (1). **ST — State Transition:** precondition valid (1), postcondition verifiable (1), state consistency (1). **SU — Scene Understanding** (global): visible objects (1), no hallucination (1), layout correct (1). **PC — Plan Consistency**

(global): no object switch (1), no contradiction (1), temporal order (1).

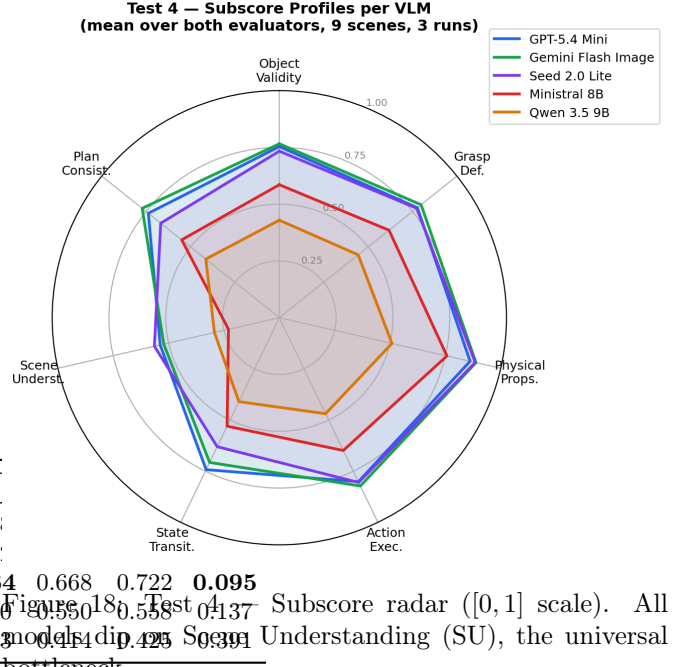
$$\text{Final} = \frac{\overline{OV} + \overline{GD} + \overline{PPC} + \overline{AE} + \overline{ST} + \overline{SU} + \overline{PC}}{7}$$

9.3 Results

Table 6 reports mean scores averaged over 10 scenes, 3 runs, and 2 evaluators. The ranking reshuffles significantly.

Table 6: Test 4 — mean subscores and final score (scores $\in [0, 1]$). σ = run std. Best per column in **bold**.

Model	OV	GD	PPC	AE	ST	SU
GPT-5.4 Mini	0.753	0.777	0.861	0.801	0.743	0.53
Gemini Flash Image	0.766	0.798	0.888	0.822	0.707	0.52
Seed 2.0 Lite	0.733	0.773	0.885	0.805	0.630	0.564
Ministral 8B	0.585	0.618	0.757	0.649	0.530	0.236
Qwen 3.5 9B	0.429	0.444	0.509	0.469	0.410	0.293



Gemini Flash Image surpasses GPT-5.4 Mini. For the first time, Gemini Flash Image takes the top position ($\mu = 0.751$) ahead of GPT-5.4 Mini ($\mu = 0.745$), leading on OV, GD, and AE. GPT-5.4 Mini compensates with a stronger State Transition score (0.743 vs 0.707), reflecting better causal reasoning about sequential action effects.

Qwen 3.5 9B shows severe degradation. Qwen 3.5 9B shows the worst performance ($\mu = 0.425$) with extremely high run variance ($\sigma = 0.391$). The *bathroom* scene produces a JSON failure again (score = 0.00). Even excluding the bathroom scene, Qwen 3.5 9B has the worst scores and variance across all remaining scenes.

Scene Understanding is the universal bottleneck. SU scores are low for all models (range 0.23–0.56), identifying global scene description as the hardest component of multi-step planning for current VLMs.

Figure 18: Test 4 — Subscore radar ([0, 1] scale). All models dip at Scene Understanding (SU), the universal bottleneck.

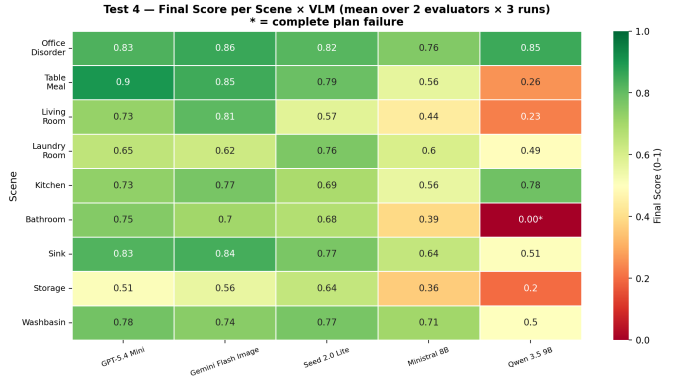


Figure 19: Test 4 — Score heatmap per scene $\times$ VLM. * = complete plan failure (Qwen 3.5 9B, bathroom).

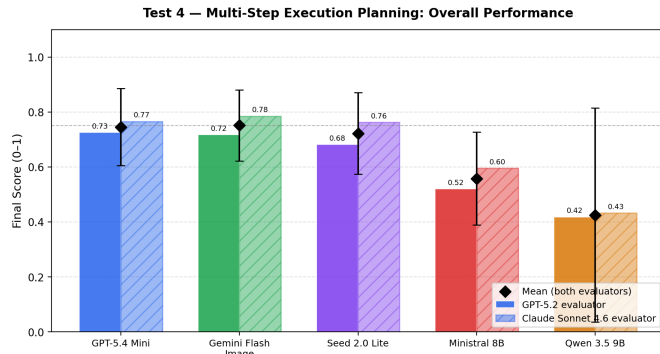


Figure 17: Test 4 — Final scores per VLM (error bars = run std). Gemini Flash Image outperforms GPT-5.4 Mini for the first time.

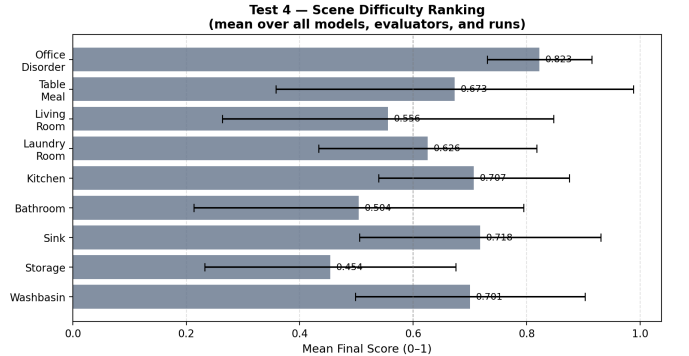


Figure 20: Test 4 — Scene difficulty ranking (mean over all models and runs).

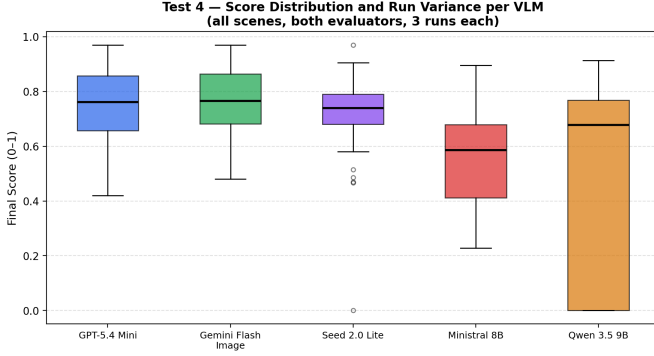


Figure 21: Test 4 — Score distribution per VLM. Qwen 3.5 9B shows extremely high variance.

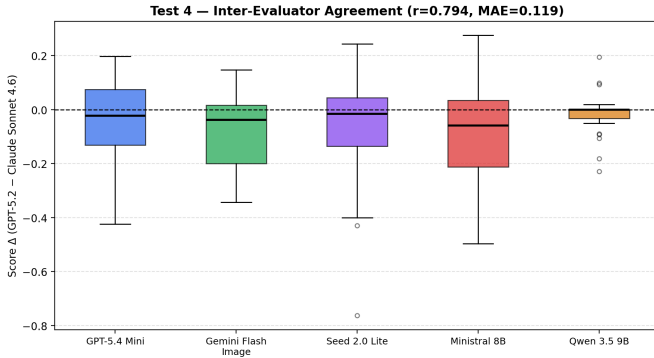


Figure 22: Test 4 — Inter-evaluator agreement ($r = 0.794$, $MAE = 0.119$).

Test 4 grounds planning in an explicit task list. The next test removes this scaffolding and forces the model to construct its own task hierarchy autonomously.

10 Test 5: High-Level Task Planning

10.1 Task and Protocol Differences

Test 5 constitutes the most demanding evaluation in the benchmark. Where Test 4 provides an explicit ordered list of manipulation steps, Test 5 gives models only a **single high-level goal description** and requires them to autonomously decompose it into atomic sub-tasks via Hierarchical Task Network (HTN) decomposition (Phase 1) before generating a grounded execution plan (Phase 2).

A unique feature of Test 5 is the **task ambiguity dimension**: each of the 10 base scenes is tested with 5 variants ranging from highly detailed descriptions (ambiguity level 1, e.g., “*tidy the desk by clearing the keyboard area, removing the cup and mouse, organizing cables, and turning on the computer setup*”) to maximally vague instructions (level 5, e.g., “*tidy the desk*”). This yields 50 scene-task

combinations evaluated over 3 runs, totaling 750 evaluations per judge.

10.2 Scoring Protocol

Test 5 uses a four-criterion rubric scored on $[0, 0.5, 1]$. **SU (Scene Understanding)**: visible objects, spatial layout, naming consistency. **TD (Task Decomposition)**: per subtask — relevance, grounding, atomicity (no abstract verbs such as *clean*, *tidy*, *organize*), physical validity, collective coverage. **AP (Action Plan)**: 11 sub-criteria per step covering subtask linkage, object validity, grasp, contact zone, accessibility, physical properties, pre/post-conditions, and state coherence. **GC (Global Consistency)**: temporal order, state contradictions, object tracking, subtask coverage, plan achieves task.

$$\text{Final} = \frac{SU + \overline{TD} + \overline{AP} + GC}{4}$$

10.3 Results

Table 7 reports mean scores averaged over 10 scenes, 5 variants, 3 runs, and 2 evaluators.

Table 7: Test 5 — mean subscores and final score ($[0, 1]$ scale). σ = score std. Best per column in **bold**.

Model	SU	TD	AP	GC	Final	σ
GPT-5.4 Mini	0.699	0.805	0.816	0.830	0.788	0.094
Gemini Flash Image	0.685	0.817	0.828	0.810	0.785	0.086
Seed 2.0 Lite	0.634	0.769	0.758	0.749	0.726	0.202
Ministral 8B	0.429	0.659	0.618	0.626	0.578	0.147
Qwen 3.5 9B	0.390	0.503	0.493	0.494	0.464	0.386

GPT-5.4 Mini and Gemini Flash Image at parity.

For the second consecutive test, GPT-5.4 Mini ($\mu = 0.788$) and Gemini Flash Image ($\mu = 0.785$) are statistically indistinguishable. GPT-5.4 Mini leads on SU and GC, while Gemini leads on TD and AP—a complementary profile suggesting different strengths: global scene interpretation vs. atomic step generation.

Scene Understanding is the universal bottleneck.

SU scores (max 0.699) are the lowest subscores for all top models, consistent with Test 4. Even the best models fail to fully enumerate visible objects and their spatial relations in roughly 30% of cases.

Qwen 3.5 9B is unreliable. Qwen 3.5 9B ($\mu = 0.464$, $\sigma = 0.386$) shows near-random performance, confirming the collapse pattern observed in Tests 2 and 4.

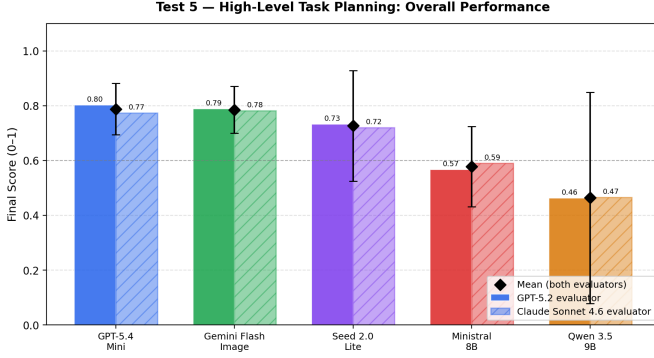


Figure 23: Test 5 — Final scores per VLM. GPT-5.4 Mini and Gemini Flash Image are at near-parity; Qwen 3.5 9B shows extremely high variance.

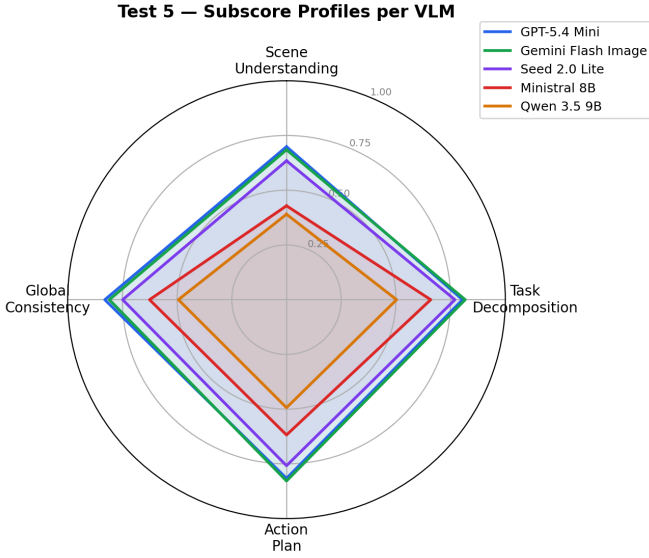


Figure 24: Test 5 — Subscore profiles. SU is the universal bottleneck; TD and AP are stronger across all models.

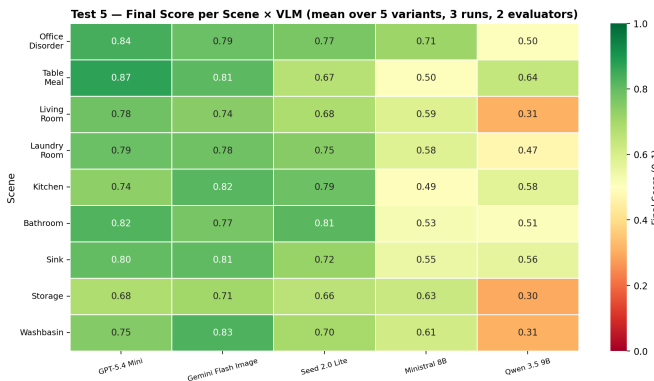


Figure 25: Test 5 — Score heatmap per base scene $\times$ VLM. The storage and sink scenes are the hardest across all models.

10.4 Task Ambiguity Analysis

Figure 26 reveals a counter-intuitive finding: **performance increases with task ambiguity** (level 1 $\mu = 0.652 \rightarrow$ level 5 $\mu = 0.714$). Detailed task descriptions impose strict coverage requirements that are harder to satisfy, while vague instructions allow models to select only the sub-tasks they are most confident about. This suggests that the TD5 criterion (*subtasks collectively cover the full task*) is easier to satisfy when the task is underspecified. Benchmark designers should account for this coverage-ambiguity trade-off when designing evaluation sets.

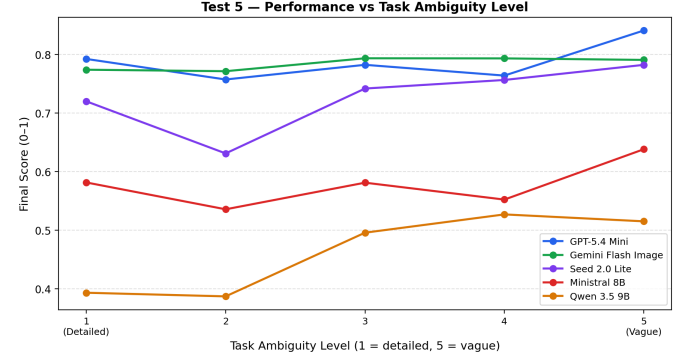


Figure 26: Test 5 — Score vs task ambiguity level. Scores generally *increase* with ambiguity—an important coverage bias to account for in benchmark design.

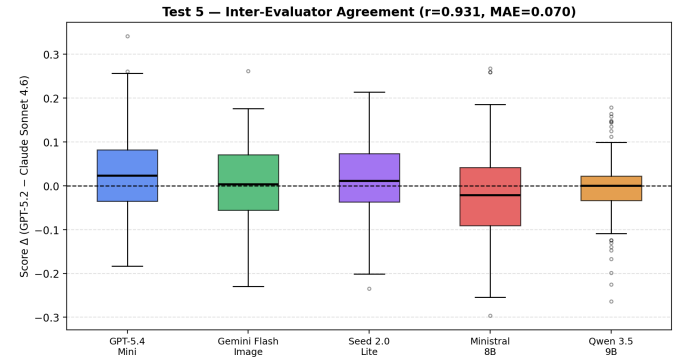


Figure 27: Test 5 — Inter-evaluator agreement ($r = 0.931$, $MAE = 0.070$): the strongest agreement across the entire benchmark.

Inter-evaluator agreement. Test 5 achieves $r = 0.931$, $MAE = 0.070$ —the best agreement across all five tests. The 0/0.5/1 scoring scale with explicit semantic definitions produces highly consistent LLM-judge evaluations.

11 Cross-Test Synthesis

Table 8 summarizes the complete benchmark ranking across all five tests, with Tests 1–3 normalized to $[0, 1]$

for comparability.

Table 8: Global benchmark: normalized final scores across all five tests (Tests 1–3 divided by 5; Tests 4–5 native [0, 1]). T1 updated with 3-run averaging.

Model	T1	T2	T3	T4	T5	Mean
GPT-5.4 Mini	0.976	0.960	0.947	0.745	0.788	0.883
Gemini Flash Image	0.934	0.914	0.866	0.751	0.785	0.850
Seed 2.0 Lite	0.920	0.932	0.922	0.722	0.726	0.844
Ministral 8B	0.782	0.768	0.666	0.558	0.578	0.670
Qwen 3.5 9B	0.856	0.704	0.786	0.425	0.464	0.647

(1) **GPT-5.4 Mini leads overall** (mean 0.883, 95% CI on T1 [4.80/5, 4.94/5]) with the most consistent cross-test performance and the lowest within-evaluator run variance ($\bar{\sigma}_{\text{run}} = 0.08$).

(2) **Gemini’s planning strength.** Gemini Flash Image remains competitive in perception-oriented tasks (2nd in Test 1, 3rd in Tests 2–3) but becomes joint-best in planning-heavy Tests 4–5. Its strengths in task decomposition and action execution compensate for weaker spatial localization performance. The updated 3-run Test 1 score (0.934 vs. original 0.926) reinforces its position above Seed 2.0 Lite overall.

(3) **Seed 2.0 Lite: the stable all-rounder.** Never placing below 3rd across all five tests, Seed 2.0 Lite is the most reliable choice for diverse deployment scenarios, despite wider confidence intervals in Test 1 ($\bar{\sigma}_{\text{run}} = 0.39$) indicating some scene-level instability.

(4) **The open-weight collapse.** Qwen 3.5 9B and Ministral 8B decline monotonically from Tests 1–3 into Tests 4–5. Both models score above 0.75 in Test 1 but fall below 0.58 by Test 5, confirming that planning complexity exposes fundamental limitations of smaller open-weight VLMs.

12 Discussion

Task format drives model ranking. Model rankings are *not stable* across tests: Gemini Flash Image is 2nd in Test 1, 3rd in Tests 2–3, and 1st in Test 4. No single test is sufficient to characterize a VLM for robotics—multi-task benchmarking is essential.

Scene Understanding as the universal bottleneck. SU is the weakest criterion in Test 4 for all models (max 0.564). This echoes the Robustness weakness in Tests 1–3: holistic scene comprehension is harder than step-level reasoning for current VLMs. Even the top models fail on roughly 30% of scene-level enumeration tasks in Test 5.

Reliability matters for robotics. Qwen 3.5 9B’s run variance ($\sigma = 0.391$) means a robot could receive wildly different plans for identical inputs. GPT-5.4 Mini and

Gemini Flash Image ($\sigma \approx 0.11$) offer acceptable consistency for deployment.

Evaluator agreement across scales. Inter-evaluator agreement improves as evaluation protocols become more structured (from $r = 0.91$ in Test 1 to $r = 0.931$ in Test 5) and as more runs per judge are available. Test 5 achieves the strongest agreement, indicating that explicit step-level semantic criteria substantially reduce scoring subjectivity. This trend supports the use of fine-grained structured rubrics for future robotics-oriented VLM benchmarks [11].

Limitations. *Dataset scope.* Tests 1–3 use 8 controlled objects and Tests 4–5 use 10 indoor scenes. Outdoor, dynamic, and occlusion-heavy environments are not represented, and no test exercises closed-loop sensorimotor feedback. Future benchmarks should extend to real robotic execution to validate whether high-scoring plans translate to successful manipulation.

LLM-as-judge bias. Both judges are themselves general-purpose LLMs whose evaluations may reflect their own visual-grounding limitations. GPT-5.2 consistently scores higher than Claude Sonnet 4.6 (mean offset +0.12 in Test 1, +0.18 in Test 2; cf. Figure 10), a leniency bias we cannot fully disentangle from the rubric.

Self-preference bias. GPT-5.2 evaluates a sibling proprietary model (GPT-5.4 Mini); the literature documents non-zero self-preference in LLM judges [11]. Although our dual-judge protocol mitigates this, it does not eliminate the risk. The consistent top ranking of GPT-5.4 Mini should therefore be interpreted with appropriate caution.

Cultural and geographic bias. The domestic scenes in Tests 4–5 predominantly depict Western kitchens, laundries, and desks. Object names, manipulation conventions, and the semantics of high-level instructions (e.g., “tidy” vs. similar concepts in other cultures) are culturally loaded and may systematically advantage models trained on Western-dominant data. Future benchmarks should incorporate geographically diverse scenes.

Scoring instability and single-run bias. Run-to-run variance is non-negligible even for proprietary models (Test 1 $\bar{\sigma}_{\text{run}} = 0.08$; Test 4 $\sigma = 0.10$ –0.14). Single-run benchmarks systematically under-represent this instability. Our 3-run protocol for Test 1 demonstrates that single-run inter-evaluator agreement ($r = 0.50$) can underestimate the true reliability by $\Delta r \approx 0.41$, motivating multi-run LLM-as-judge protocols as a methodological standard for future open-ended evaluations. Tests 2–3 remain single-run and should be updated accordingly in future work.

Rubric subjectivity. Open-ended subscores (notably Robustness in Tests 1–3 and Scene Understanding in Tests 4–5) carry residual subjectivity even with structured criteria, as evidenced by the lower Test 1 agreement relative to Test 5 ($r = 0.91$ vs. $r = 0.93$).

13 Conclusion

We presented a five-test benchmark progressively scaling from single-object perception to autonomous hierarchical planning, evaluated with two LLM judges across three independent runs in Test 1. The benchmark exposes a strict capability hierarchy: *proprietary VLMs reach near-ceiling on perception but lose 15–25% of their normalized score under planning load, while open-weight VLMs collapse on spatial grounding*. Scene Understanding is the universal bottleneck and the only criterion below 0.6 for every model in Tests 4–5. Run-level variance (Test 4: $\sigma = 0.39$ for Qwen 3.5 9B) disqualifies the weakest open-weight model for safety-critical deployment.

Multi-run LLM-as-judge protocols are essential: single-run agreement underestimates true inter-judge reliability by up to $\Delta r \approx 0.41$ in open-ended settings, but converges toward structured-task agreement ($r \approx 0.91$) once three runs are averaged per judge. Future work should validate top-ranked plans on real robots and incorporate human expert ratings as a third evaluation channel to ground LLM-judge scores in physical outcomes.

Acknowledgments

The author thanks his internship supervisors Anany Dwivedi, Sudhir Zhuwawu and Nick Lim for their rigorous guidance, sound advice, and availability throughout this research project. Their scientific expertise and methodological guidance greatly contributed to the progress of this work and to deepening the author’s knowledge in the field of machine learning.

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